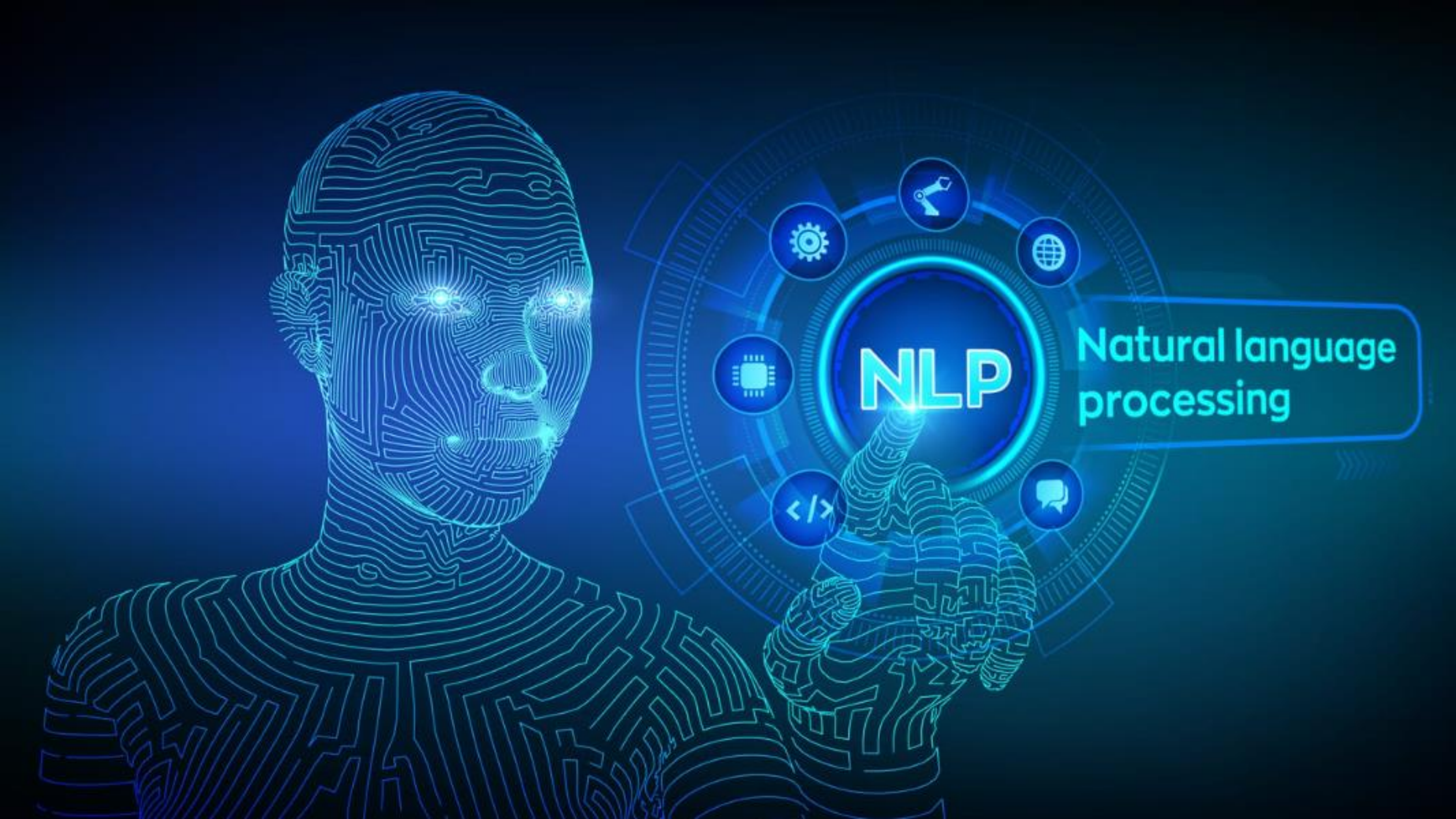

NLP
Natural language processing

Prerequisite
for
Natural language processing

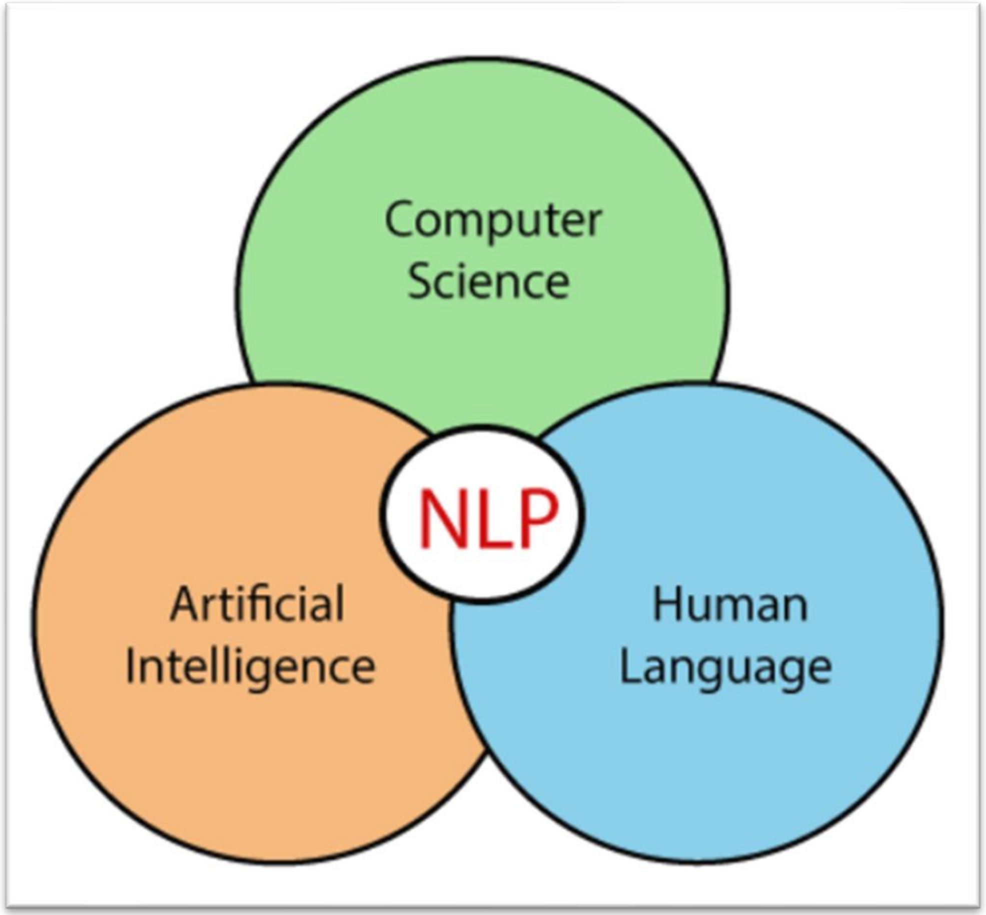
- ***Python***
- ***Basic Concept of Machine Learning and Deep Learning***

Natural language processing

Natural language processing (NLP) is a subfield of computer science, information engineering, and artificial intelligence concerned with the interactions between computers and human (natural) languages, in particular how to program computers to process and analyze large amounts of natural language data.

Challenges in natural language processing frequently involve speech recognition, natural language understanding, and natural language generation.

It helps developers to organize knowledge for performing tasks such as **translation, automatic summarization, Named Entity Recognition (NER), speech recognition, relationship extraction,** and **topic segmentation**.



•**Translation**: Automatically converting text or speech from one language to another, like translating English sentences into French.

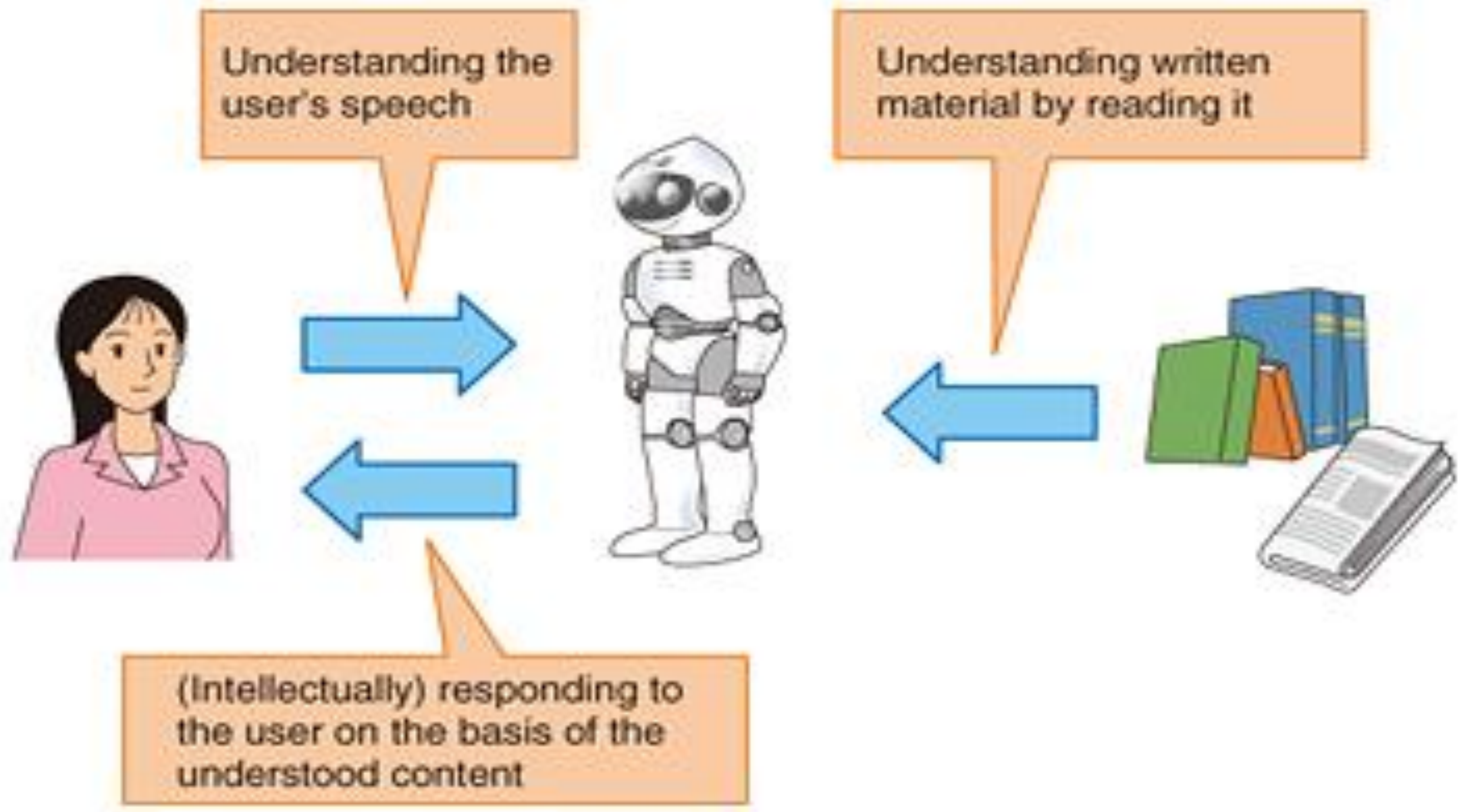
•**Automatic Summarization**: Generating a condensed version of a longer text while retaining its key points. It can be **extractive** (picking important sentences directly) or **abstractive** (creating new sentences that summarize the text).

•**Named Entity Recognition (NER)**: Identifying and classifying entities (like names of people, organizations, dates, etc.) in text. For example, in the sentence "John works at Microsoft," NER would label "John" as a person and "Microsoft" as an organization.

•**Speech Recognition**: Converting spoken language into text. It's what happens when you use voice assistants like Siri or Google Assistant.

•**Relationship Extraction**: Identifying relationships between entities in a text. For example, in "John works at Microsoft," this technique would extract the relationship "works_at" between "John" and "Microsoft."

•**Topic Segmentation**: Dividing a text into segments, each dealing with a specific topic. This is useful in longer documents, like news articles or research papers, to identify different themes or subjects.


Understanding the user's speech
Understanding written material by reading it
(Intellectually) responding to the user on the basis of the understood content

Applications of NLP

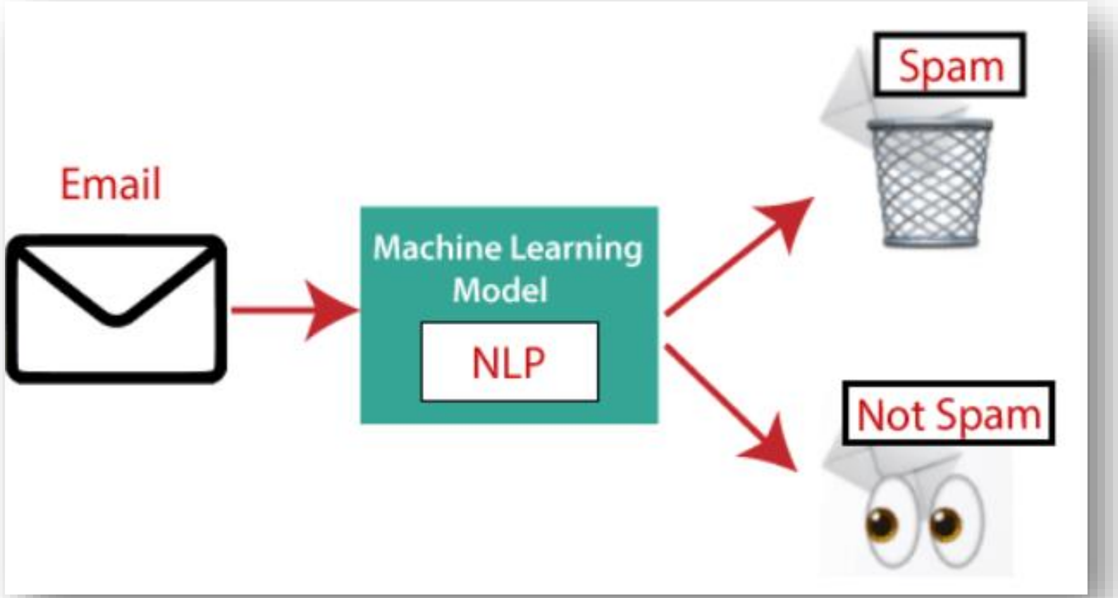
There are the following applications of NLP -

1. Smart Assistants

Smart Assistants focuses on building systems that automatically answer the questions asked by humans in a natural language.

2. Spam Detection

Spam detection is used to detect unwanted e-mails getting to a user's inbox.



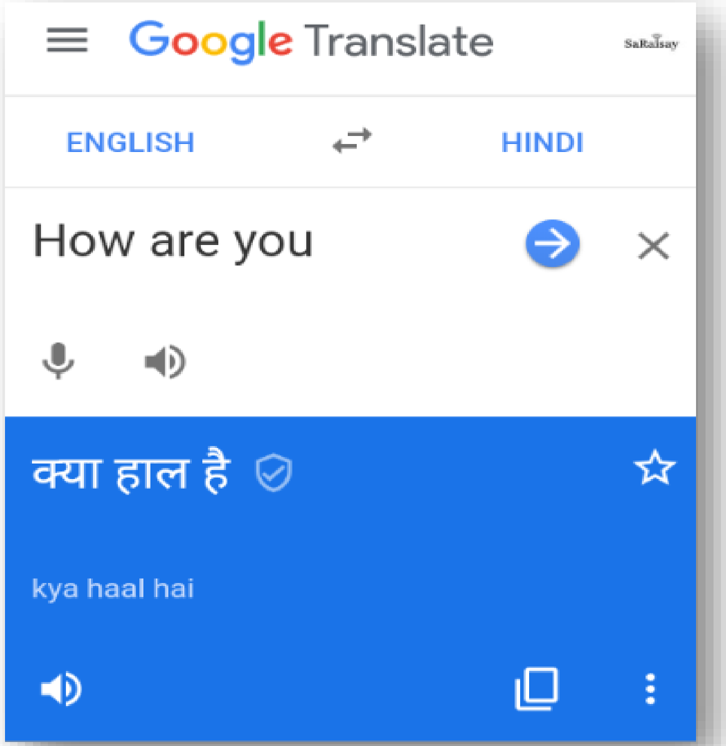
3. Sentiment Analysis

Sentiment Analysis is also known as **opinion mining**. It is used on the web to analyse the attitude, behaviour, and emotional state of the sender. This application is implemented through a combination of NLP (Natural Language Processing) and statistics by assigning the values to the text (positive, negative, or natural), identify the mood of the context (happy, sad, angry, etc.)

4. Language Translation

Language translation is used to translate text or speech from one natural language to another natural language.

Example: Google Translator

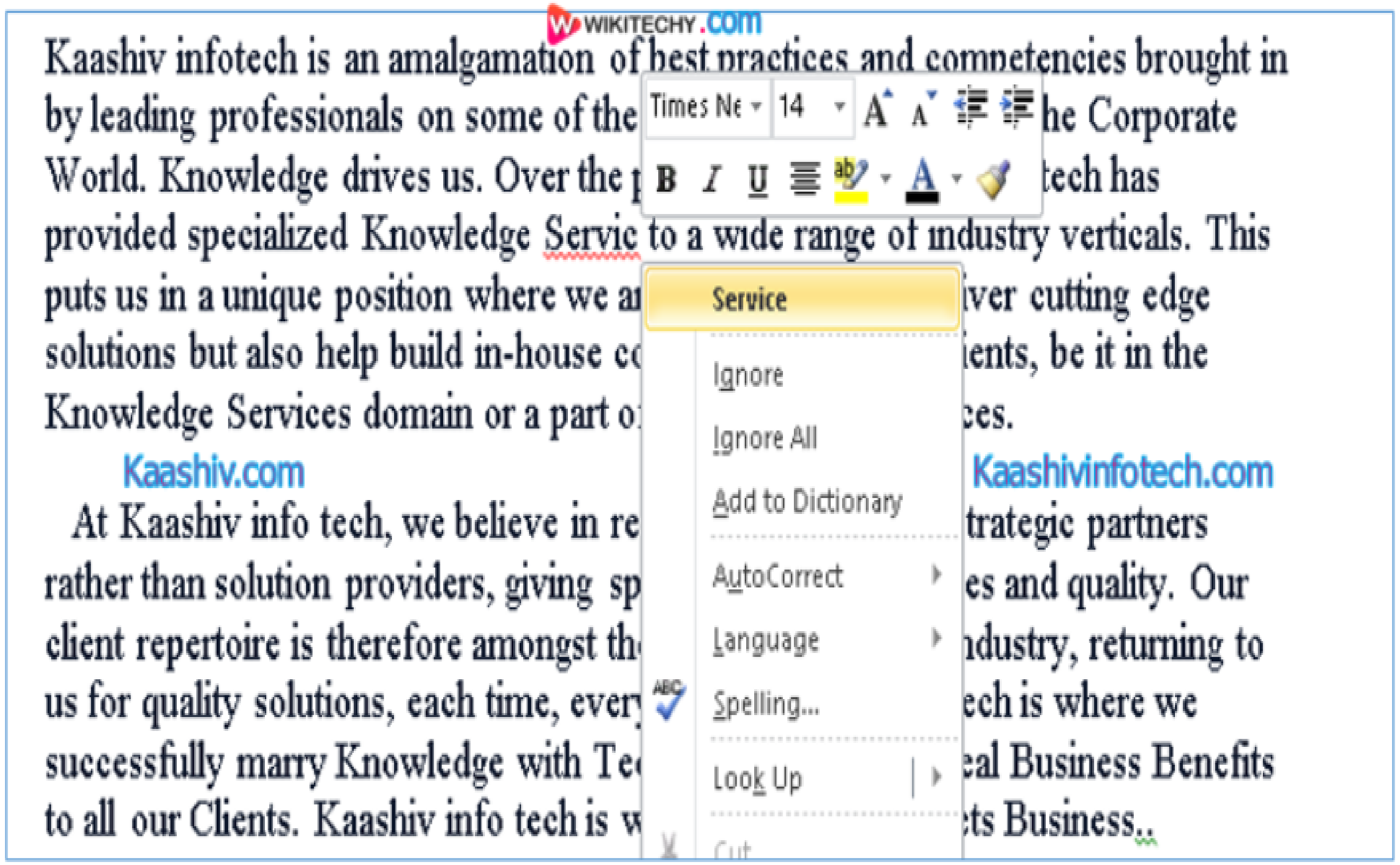


5. Spelling correction

Microsoft Corporation provides word processor software like MS-word, PowerPoint for the spelling correction.

6. Speech Recognition

Speech recognition is used for converting spoken words into text. It is used in applications, such as mobile, home automation, video recovery, dictating to Microsoft Word, voice biometrics, voice user interface, and so on.



7. Chatbot

Implementing the Chatbot is one of the important applications of NLP. It is used by many companies to provide the customer's chat services.

8. Information extraction

Information extraction is one of the most important applications of NLP. It is used for extracting structured information from unstructured or semi-structured machine-readable documents.

Applications

Speech
Transcription

Neural Machine
Translation (NMT)

Chatbots

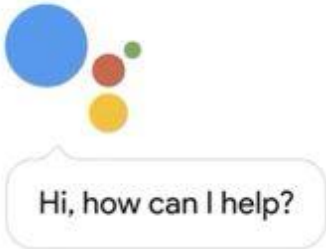
Q&A

Text
Summarization

Image
Captioning

Video
Captioning

Used by



Advantages of NLP

- NLP helps users to ask questions about any subject and get a direct response within seconds.
- NLP offers exact answers to the question means it does not offer unnecessary and unwanted information.
- NLP helps computers to communicate with humans in their languages.
- It is very time efficient.
- Most of the companies use NLP to improve the efficiency of documentation processes, accuracy of documentation, and identify the information from large databases.

Disadvantages of NLP

- For the training of the NLP model, A lot of data and computation are required.
- Many issues arise for NLP when dealing with informal expressions, idioms, and cultural jargon.
- NLP results are sometimes not to be accurate, and accuracy is directly proportional to the accuracy of data.
- NLP is designed for a single, narrow job since it cannot adapt to new domains and has a limited function.

Natural Language Processing (NLP): Classical vs. Deep

Natural Language Processing (NLP) is a field of Artificial Intelligence that enables computers to understand, interpret, and generate human language. Two main approaches have emerged: Classical NLP and Deep NLP.

What is Classical NLP?

Rule-Based

Classical NLP relies on hand-crafted rules and patterns to analyze and process text.

Symbolic Representation

It represents language as a system of symbols and structures, enabling logical reasoning.

Classical NLP

•**Overview**: Classical NLP uses rule-based, statistical, and machine learning methods to process language. These approaches were dominant before the rise of deep learning techniques.

•**Key Features**:

- Focus on **linguistic rules**: Uses grammar, syntax rules, and domain-specific dictionaries.
- Heavy reliance on **handcrafted features** and **manual feature engineering**.
- Often employs **shallow machine learning models** such as:
  - **Naive Bayes**
  - **Support Vector Machines (SVM)**
  - **Hidden Markov Models (HMM)**
  - **Conditional Random Fields (CRF)**
- Models like **n-grams**, **Bag of Words (BoW)**, and **TF-IDF** (Term Frequency-Inverse Document Frequency) are commonly used for text representation.

•**Examples**:

- Spam classification using TF-IDF with an SVM classifier.
- Named Entity Recognition (NER) using rule-based methods or CRFs.
- POS tagging using HMMs or CRFs.

•**Limitations**:

- Requires **extensive feature engineering**.
- Not very good at capturing long-term dependencies or the contextual meaning of words.
- Struggles with large datasets and complex tasks like machine translation or sentiment analysis at scale.

What is Deep NLP?

1

Neural Networks

Deep NLP uses artificial neural networks to learn complex patterns and relationships in language data.

2

Data-Driven

These models learn from large amounts of text data, adapting to the nuances and complexities of human language.

3

End-to-End Learning

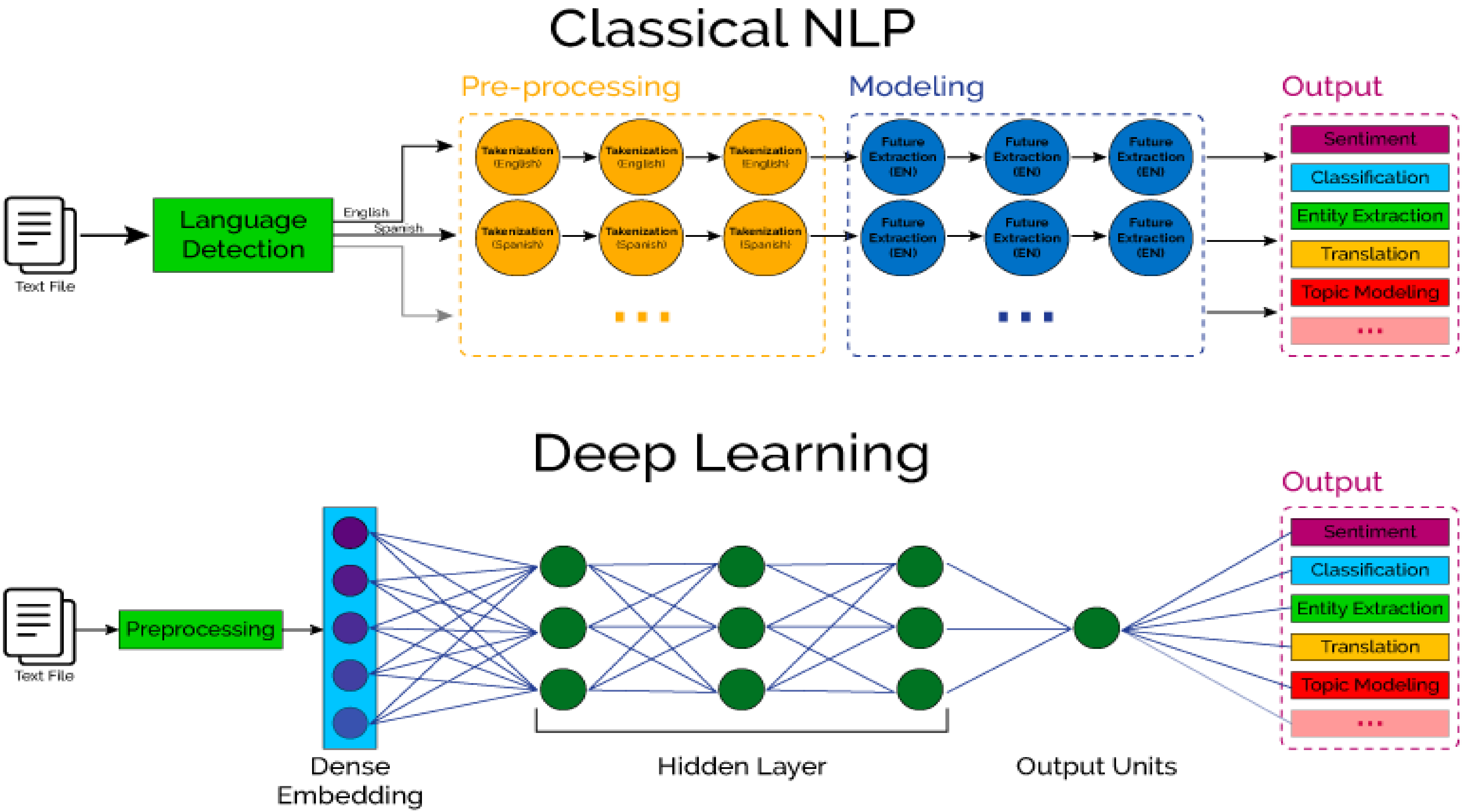
Deep NLP models can perform multiple NLP tasks in a single pipeline, achieving greater accuracy and efficiency.

Deep NLP (Deep Learning-based NLP)

•**Overview**: Deep NLP leverages neural networks, particularly deep learning architectures, to learn patterns in language data without requiring handcrafted features. It has gained prominence due to its superior performance in complex NLP tasks.

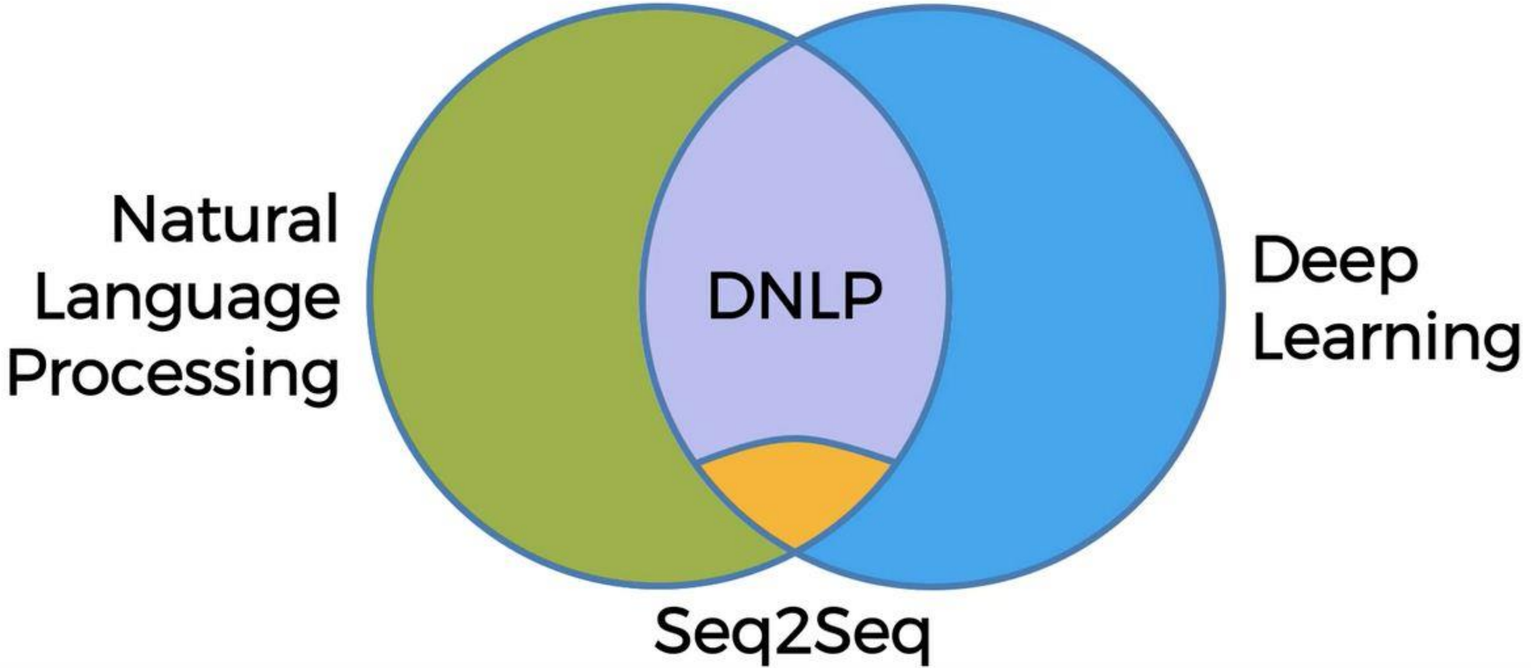
•**Key Features**:

- Uses **deep learning models** like:
  - **Recurrent Neural Networks (RNN)** and variants like **LSTMs** and **GRUs**.
  - **Convolutional Neural Networks (CNN)** (applied to text).
  - **Transformers** (e.g., BERT, GPT).
- No need for manual feature engineering—models automatically learn features from raw text data.
- Able to **capture context** and **long-term dependencies** in language.
- Often relies on **word embeddings** such as **Word2Vec**, **GloVe**, or **BERT embeddings** for text representation.



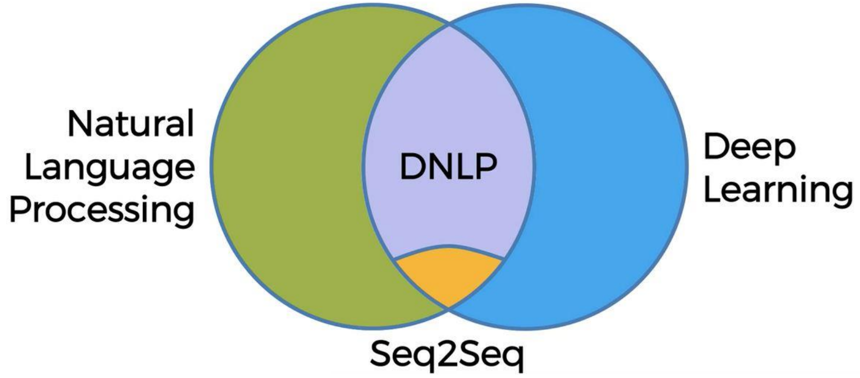
Classical NLP
Pre-processing
Modeling
Output
Text File
Language Detection
English
Spanish
Tokenization (English)
Tokenization (Spanish)
Future Extraction (EN)
Sentiment
Classification
Entity Extraction
Translation
Topic Modeling
...
Deep Learning
Output
Text File
Preprocessing
Dense Embedding
Hidden Layer
Output Units
Sentiment
Classification
Entity Extraction
Translation
Topic Modeling
...

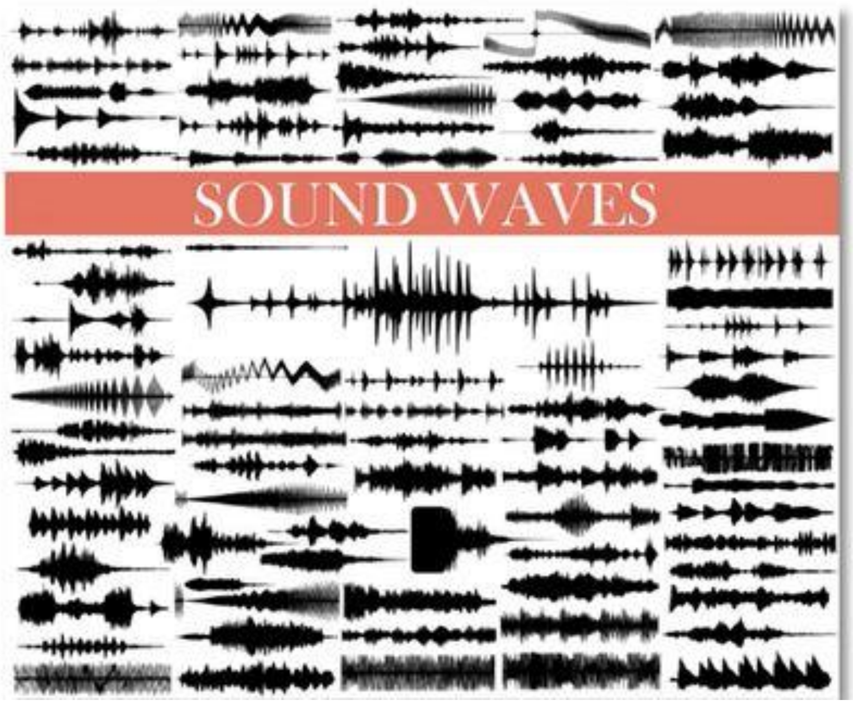
How NLP, DNLP and DL involves in!!!

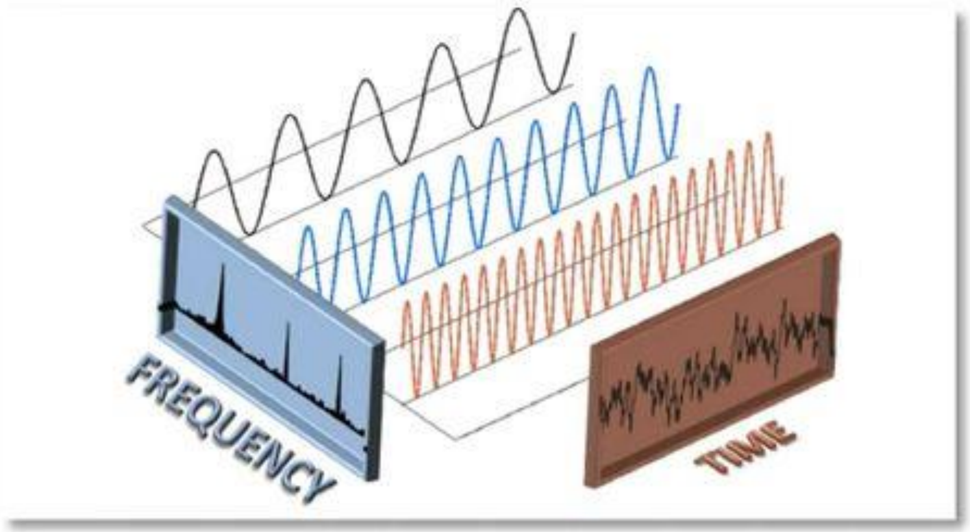


1. If / Else Rules (Chatbot)
2. Audio frequency components analysis (Speech Recognition)
3. Bag-of-words model (Classification)
4. CNN for text Recognition (Classification)
5. Seq2Seq (many applications)

How NLP, DNLP and DL involves in!!!







How NLP work?

Components of NLP

There are the following two components of NLP -

1. Natural Language Understanding (NLU)

Natural Language Understanding (NLU) helps the machine to understand and analyse human language by extracting the metadata from content such as concepts, entities, keywords, emotion, relations, and semantic roles.

NLU mainly used in Business applications to understand the customer's problem in both spoken and written language.

NLU involves the following tasks -

- It is used to map the given input into useful representation.
- It is used to analyze different aspects of the language.

2. Natural Language Generation (NLG)

Natural Language Generation (NLG) acts as a translator that converts the computerized data into natural language representation. It mainly involves Text planning, Sentence planning, and Text Realization.

NLP Working


NLP
NLG
NLU
Natural Language Processing is computers reading language
Natural Language Generation is computers writing language.
Natural Language Understanding is computers understanding language. e.g. Siri & Amazon Alexa.

Natural Language Understanding

Phonology – This science helps to deal with patterns present in the sound and speeches related to the sound as a physical entity.

Pragmatics – This science studies the different uses of language.

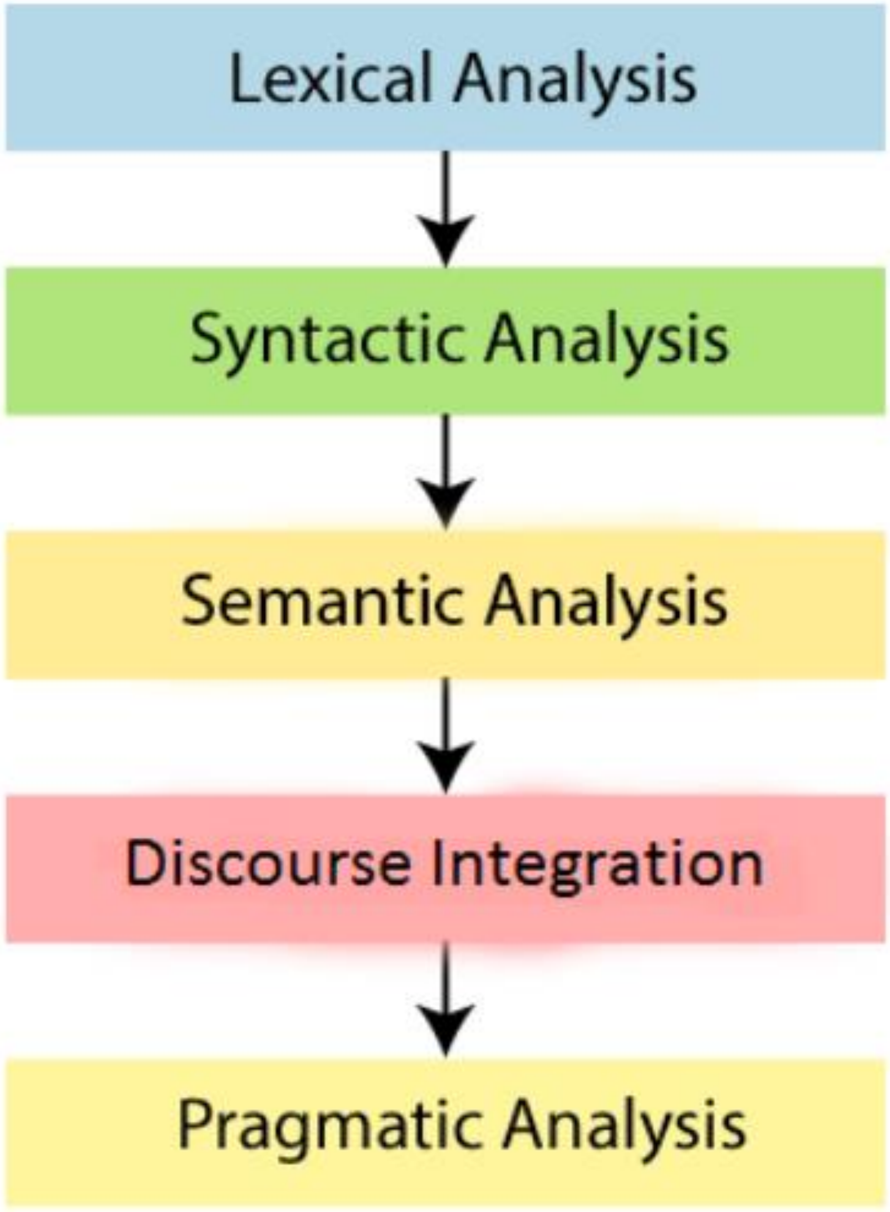
Morphology – This science deals with the structure of the words and the systematic relations between them.

Syntax – This science deal with the structure of the sentences.

Semantics – This science deals with the literal meaning of the words, phrases as well as sentences.

Phases of NLP

There are the following five phases of NLP:



1. Lexical Analysis and Morphological

The first phase of NLP is the Lexical Analysis. This phase scans the source code as a stream of characters and converts it into meaningful lexemes. It divides the whole text into paragraphs, sentences, and words.

2. Syntactic Analysis (Parsing)

Syntactic Analysis is used to check grammar, word arrangements, and shows the relationship among the words.

Example: Agra goes to the Poonam

In the real world, Agra goes to the Poonam, does not make any sense, so this sentence is rejected by the Syntactic analyzer.

3. Semantic Analysis

Semantic analysis is concerned with the meaning representation. It mainly focuses on the literal meaning of words, phrases, and sentences.

4. Discourse Integration

Discourse Integration depends upon the sentences that proceeds it and also invokes the meaning of the sentences that follow it.

5. Pragmatic Analysis

Pragmatic is the fifth and last phase of NLP. It helps you to discover the intended effect by applying a set of rules that characterize cooperative dialogues.

For Example: "Open the door" is interpreted as a request instead of an order.

Natural Language Generation

Based on NL-Understanding, it will suggest about:

- **What should say to user.**
- **Should be Intelligent and Covervational as like human**
- **Usage of Structured data.**
- **With text and Sentence like planning.**

NLP = NLU + NLG

Difference between NLU and NLG

NLU	NLG
NLU is the process of reading and interpreting language.	NLG is the process of writing or generating language.
It produces non-linguistic outputs from natural language inputs.	It produces constructing natural language outputs from non-linguistic inputs.

Difference between Natural language and Computer Language

Natural Language	Computer Language
Natural language has a very large vocabulary.	Computer language has a very limited vocabulary.
Natural language is easily understood by humans.	Computer language is easily understood by the machines.
Natural language is ambiguous in nature.	Computer language is unambiguous.

Why NLP is difficult?

NLP is difficult because Ambiguity and Uncertainty exist in the language.

Ambiguity

There are the some following ambiguity –

Ambiguity:

- **Lexical Ambiguity : The Tank is full of water.**
- **Syntactic Ambiguity : ill men and women get to hospital.**
- **Semantic Ambiguity : The Bike hit the pole while it was running.**
- **Pragmatic Ambiguity : The Army is coming.**

•**Lexical Ambiguity**

Lexical Ambiguity exists in the presence of two or more possible meanings of the sentence within a single word.

Example:

Manya is looking for a **match**.

In the above example, the word match refers to that either Manya is looking for a partner or Manya is looking for a match. (Cricket or other match)

•**Syntactic Ambiguity**

Syntactic Ambiguity exists in the presence of two or more possible meanings within the sentence.

Example:

I saw the moon with the binocular.

In the above example, did I have the binoculars? Or did the moon have the binoculars?

•**Referential Ambiguity**

Referential Ambiguity exists when you are referring to something using the pronoun.

Example: Kiran went to Sunita. She said, "I am hungry."

In the above sentence, you do not know that who is hungry, either Kiran or Sunita.

How to build an NLP pipeline

There are the following steps to build an NLP pipeline -

Step1: Sentence Segmentation

Sentence Segment is the first step for building the NLP pipeline. It breaks the paragraph into separate sentences.

Example: Consider the following paragraph -

Independence Day is one of the important festivals for every Indian citizen. It is celebrated on the 15th of August each year ever since India got independence from the British rule. The day celebrates independence in the true sense.

Sentence Segment produces the following result:

1."Independence Day is one of the important festivals for every Indian citizen."

2."It is celebrated on the 15th of August each year ever since India got independence from the British rule."

3."This day celebrates independence in the true sense."

Step2: Word Tokenization

Word Tokenizer is used to break the sentence into separate words or tokens.

Example:

They offers Corporate Training, Summer Training, Online Training, and Winter Training.

Word Tokenizer generates the following result:

“They", "offers", "Corporate", "Training", "Summer", "Training", "Online", "Training", "and", "Winter", "Training", "."

Step3: Stemming

Stemming is used to normalize words into its base form or root form. For example, celebrates, celebrated and celebrating, all these words are originated with a single root word "celebrate." The big problem with stemming is that sometimes it produces the root word which may not have any meaning.

For Example, intelligence, intelligent, and intelligently, all these words are originated with a single root word "intelligen." In English, the word "intelligen" do not have any meaning.

Step 4: Lemmatization

Lemmatization is quite similar to the Stemming. It is used to group different inflected forms of the word, called Lemma. The main difference between Stemming and lemmatization is that it produces the root word, which has a meaning.

For example: In lemmatization, the words intelligence, intelligent, and intelligently has a root word intelligent, which has a meaning.

Step 5: Identifying Stop Words

In English, there are a lot of words that appear very frequently like "is", "and", "the", and "a". NLP pipelines will flag these words as stop words. **Stop words** might be filtered out before doing any statistical analysis.

Example: He **is a** good boy.

Step 6: Dependency Parsing

Dependency Parsing is used to find that how all the words in the sentence are related to each other.

Step 7:Count vectorizer

Count Vectorizer is a way to convert a given set of strings into a frequency representation. In the above two examples you have Texts that are Tagged respectively. This is a very simple case of NLP where you get tagged text data set and then using it you have to predict the tag of another text data.

Step 8: Part-of-speech tagging (POS tags)

POS stands for parts of speech, which includes Noun, verb, adverb, and Adjective. It indicates that how a word functions with its meaning as well as grammatically within the sentences. A word has one or more parts of speech based on the context in which it is used.

Example: "Google" something on the Internet.

In the above example, Google is used as a verb, although it is a proper noun.

Step 9: Named Entity Recognition (NER)

Named Entity Recognition (NER) is the process of detecting the named entity such as person name, movie name, organization name, or location.

Example: Steve Jobs introduced iPhone at the Macworld Conference in San Francisco, California.

Step 10: Chunking

Chunking is used to collect the individual piece of information and grouping them into bigger pieces of sentences.

NLP Libraries

Scikit-learn: It provides a wide range of algorithms for building machine learning models in Python.
Natural language Toolkit (NLTK): NLTK is a complete toolkit for all NLP techniques.
Pattern: It is a web mining module for NLP and machine learning.
TextBlob: It provides an easy interface to learn basic NLP tasks like sentiment analysis, noun phrase extraction, or pos-tagging.
Quepy: Quepy is used to transform natural language questions into queries in a database query language.
SpaCy: SpaCy is an open-source NLP library which is used for Data Extraction, Data Analysis, Sentiment Analysis, and Text Summarization.
Gensim: Gensim works with large datasets and processes data streams.

1. Natural Language Toolkit (NLTK)
2. TextBlob
3. CoreNLP
4. Gensim
5. spaCy
6. polyglot
7. scikit–learn
8. Pattern